M T W T F S S

COMPASS

Date :

1. Srinivas P Bodke.
26 1100690018

Cryptology Assignment 8.

1. Determine the multiplicative inverse modulo 26 for each of the following integers

1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25 mod 26

Ans

i) multiplicative inverse of 1.

$26 = 1(26) + 0$

$1(1) = 1$

$1^{-1} = 1 \pmod{26}$

multiplicative inverse of $1 = 1$

ii) multiplicative inverse of 3

$26 = 3(8) + 2$

$3 = 2(1) + 1$

$1 = 3 - 2$ — ①

$2 = 26 - 3(8)$

$1 = 3 - (26 - 3(8))$

$1 = 3 - 26 + 3(8)$

$1 = 3(9) + 26(-1)$

$3^{-1} = 9 \pmod{26}$

multiplicative inverse of $3 = 9$.

iii) multiplicative inverse of 5

$26 = 5(5) + 1$

$1 = 26 - 5(5)$

$1 = 5(5) + 26(1)$

$-5 + 26 = 21$

$5^{-1} = 21 \pmod{26}$

multiplicator inverse of $5 = 21$

iv) multiplicative inverse of 7

$26 = 7(3) + 5$, $5 = 26 - 7(3)$

$7 = 5(1) + 2$, $2 = 7 - 5$ — ②.

$5 = 2(2) + 1$

$1 = 5 - 2(2)$ — Sub ② in ①

$1 = 5 - 2\times 7 + 2\times 5$

$1 = 3(5) - 2(7)$

$1 = 3(26 - 7(3)) - 2(7)$

$1 = 3\times 26 - 3\times(7(3)) - 2(7)$

$1 = 3\times 26 - 11\times 7$

$1 = (-11)\times 7 - 3\times 26$, $-11 + 26 = 15$.

$7^{-1} = 15 \pmod{26}$

multiplicative inverse of 7 is 15.

v) multiplicative inverse of 9.

$26 = 9(2) + 8$

$9 = 8(1) + 1$

$1 = 9 - 8$, $8 = 26 - 9(2)$

$1 = 9 - (26 - 9(2))$

$1 = 9 + 9(2) - 26$

$1 = 3(9) - 26$

$9^{-1} = 3 \pmod{26}$

multiplicative inverse of 9 is 3.

vi) multiplicative inverse of 11

$26 = 11(2) + 4$

$11 = 4(2) + 3$, $3 = 11 - 4(2)$

$4 = 3(1) + 1$, $4 = 26 - 11(2)$

$1 = 4 - 3$

$1 = 4 - (11 - 4(2))$ $= 4 - 11 + 4(2)$

$1 = 3(4) - 11$

M T W T F S S

Date :

$1 = 3(26 - 11(2)) - 11$

$1 = 3 \times 26 - 3 \times (11(2)) - 11$

$1 = 3 \times 26 - 7 \times 11$

$1 = (-7)11 + 3 \times 26$, $-7 + 26 = 19$

$11^{-1} = 19 \mod (26)$

multiplicate inverse of 11 is 19

viii) Inverse of 15,

$26 = 15(1) + 11$, $11 = 26 - 15$

$15 = 11(1) + 4$, $4 = 15 - 11(1)$

$11 = 4(2) + 3$, $3 = 11 - 4(2)$

$4 = 3(1) + 1$

$1 = 4 - 3$

$1 = 4 - (11 - 4(2))$

$1 = 3(4) - 11$

$1 = 3(15 - 11(1)) - 11$

$1 = 3 \times 15 - 4 \times 11$

$1 = 3 \times 15 - 4(26 - 15)$

$1 = 7(15) - 4(26)$

$15^{-1} = 7 \ (\mod 26)$

multiplicative inverse of 15 is 7

vii) Inverse of 17,

$26 = 17(1) + 9$, $9 = 26 - 17(1)$

$17 = 9(1) + 8$, $8 = 17 - 9(1)$

$9 = 8(1) + 1$

$1 = 9 - 8$

$1 = 9 - (17 - 9(1))$

$1 = 2(9) - 17$

$1 = 2(26 - 17(1)) - 17$

$1 = 2 \times 26 - 2 \times (17(1)) - 17$

$1 = 2 \times 26 - 13 \times (17)$

M T W T F S S

COMPASS

Date :

$26 - 3 \leftarrow 1 = (-3)17 - 2(26)$

23 multiplicative inverse of 17 is 23

ix) Inverse of 19,

$26 = 19(1) + 7$, $7 = 26 - 19$.

$19 = 7(2) + 5$; $5 = 19 - 7(2)$

$7 = 5(1) + 2$, $2 = 7 - 5(1)$.

$5 = 2(2) + 1$

$1 = 5 - 2(2)$

$1 = 5 - 2(7 - 5(1))$

$1 = 5 - 2 \times 7 + 2(5(1))$

$1 = 3(5) - 2(7)$

$1 = 3(19 - 7(2)) - 2(7)$

$1 = 3 \times 19 - 8(7)$

$1 = 3(19) - 8(26 - 19)$

$1 = 11(19) - 8(26)$

$19^{-1} = 11 \bmod 26$

multiplicative inverse of 19 is 11

x) Inverse of 21,

$26 = 21(1) + 5$, $5 = 26 - 21$

$21 = 5(4) + 1$

$1 = 21 - 5(4)$

$1 = 21 - 4(26 - 21)$

$1 = 21 - 4(26) + 4(21)$

$1 = 5(21) - 4(26)$

$21^{-1} = 5 \bmod 26$.

multiplicative inverse of 21 is 5.

M T W T F S S

Date :

(x i) Inverse of 23,

$26 = 23(1) + 3$, $3 = 26 - 23$

$23 = 3(7) + 2$. $2 = 23 - 3(7)$.

$3 = 2(1) + 1$

$1 = 3 - 2(1)$

$1 = 3 - (23 - 3(7))$

$1 = 8(3) - 23$

$1 = 8(26 - 23) - 23$

$1 = 8(26 - 23) - 23$

$1 = (-9)23 + 26(8)$

$26 - 9 = 17$

$23^{-1} = 17 \mod 26$

multiplicative inverse of 23 is 17.

(x ii) Inverse of 25,

$26 = 25(1) + 1$

$1 = 26 - 25$

$1 = (-1)25 + 26$

$26 - 1 = 25$

$25^{-1} = 25 \pmod{26}$

Inverse of 25 is 25.

2. Choose any string of your choice to encrypt using the offline cipher with key values multiplier 5 & Addictive 8.

Ans:- For Encryption.

$p\ (CK_1 * P + K_2) \mod 26$

given

$K_1 = 5$, $K_2 = 8$

M T W T F S S

COSMOS

Date :

plaintext		$K_1 \times P$		$+K_2$	mod 26	
G	6	5×6	$= 30$	38	12	M
O	14	5×14	$= 70$	78	0	a
O	14	5×14	$= 70$	78	0	a
D	3	5×3	$= 15$	23	23	x
L	11	5×11	$= 55$	63	11	L
U	20	5×20	$= 100$	108	4	e
C	2	5×2	$= 10$	18	18	S
K	10	5×10	$= 50$	58	6	g

cipher text : MAAXLESG

3. Encrypt "HI" using the Affine cipher with $a = 11$ & $b = 5$

Ans. for Encryption:

$$C = E(K_1 * P + K_2) \bmod 26$$

$$K_1 = 11, \quad K_2 = 5$$

plain txt		$K_1 \times P$		$+K_2$	mod 26		cipher
H	7	11×7	$= 77$	$77 + 5$	$= 82$	4	E
I	8	11×8	$= 88$	$88 + 5$	$= 93$	15	P

cipher text : EP

4. The ciphertext UCR was Encrypted using the affine function $9x + 2 \bmod 26$ find the plain text.

Ans. for Decryption:

$$P = K^{-1}(C - K_2) \bmod 26$$

$$K_1 = 9 \qquad K_2 = 2.$$

$$9^{-1} \bmod 26$$

$$26 = 9(2) + 8, \quad 8 = 26 - 9(2).$$

$$9 = 8(1) + 1$$

$$1 = 9 - 8$$

$$1 = 9 - (26 - 9(2))$$

$$1 = 3(9) - 26 \qquad 9^{-1} \bmod 26 = 3$$

COMPASS

M T W T F S S

Date :

ciphertext		$k_1^{-1}(c-k_2)$	mod 26	Plaintext
U	20	$3(20-2) = 54$	2	c
C	2	$3(2-2) = 0$	0	a
R	17	$3(17-2) = 45$	19	t

plaintext : CAT

5. Decrypt the cipher text using Affine cipher, KDYXJUWDEXZY using the key : 3 (mul) & 4.

Ans for decryption.

$$p = k_1^{-1} \cdot (c - k_2) \bmod 26.$$

Cipher text : KDYXJUWDEXZY

$k_1 = 3$ $k_2 = 4$

$3^{-1} \bmod 26$

$26 = 3(8) + 2$, $2 = 26 - 3(8)$

$3 = \cancel{1}\,2(1) + 1$

$1 = 3 - 2$

$1 = 3 - (26 - 3(8))$

$1 = 3(9) - 26$

$3^{-1} \bmod 26 = 9$

M T W T F S S

COMPASS

Date :

cipher text		$k^{-1}(c - k_2)$		mod 26	plaintxt.
K	10	9 (10 - 4)	= 54	2	C
D	3	9 (3 - 4)	= -9	17	R
Y	24	9 (24 - 4)	= 180	24	Y
X	23	9 (23 - 4)	= 171	15	P
J	9	9 (9 - 4)	= 45	19	T
U	20	9 (20 - 4)	= 144	14	O
W	22	9 (22 - 4)	= 162	6	G
D	3	9 (3 - 4)	= -9	17	R
E	4	9 (4 - 4)	= 0	0	A
X	23	9 (23 - 4)	= 171	15	P
Z	25	9 (25 - 4)	= 189	7	H
Y	24	9 (24 - 4)	= 180	24.	Y.

plaintxt :- CRYPTOGRAPHY.

6. Use k_1 = 13 (multiplier) k_2 = 4 (Even though it not a valid key try to Encrypt). encrypt the string " input " & " alter " using the above key & find the what is issue.

Ans :- for Encryption ; $C = E\,(k_1 * P + k_2) \bmod 26$.

$k_1 = 13 \quad k_2 = 4.$

plain text		$k_1 * P$		$+ k_2$.			
I	8	13 x 8	= 104	104 + 4	= 108	4	E
N	13	13 x 13	= 169	169 + 4	= 173	17	R
P	15	13 x 15	= 195	195 + 4	= 199	17	R
U	20	13 x 20	= 260	260 + 4	= 264	4	E
T	19	13 x 19	= 247	247 + 4	= 251	17	R
A	0	13 x 0	= 0	0 + 4 =	4	4	E
L	11	13 x 11	= 143	143 + 4	= 147	17	R
T	19	13 x 19	= 247	247 + 4	= 251	17	R
E	4	13 x 4	= 52	52 + 4	= 56	4	E
R	17	13 x 17	= 221	221 + 4.	= 225	17	R

M T W T F S S

Date :

$K_1 = 13$ is not coprime with 26.
The entire 26 letter alphabet is compressed into just two ciphertext letter (E & R)

7. Find the Inverse Matrix for the key.

$$\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \text{ mod } 26.$$

$= 3\times 7 - 5\times 2 = 21 - 10 = 11$

11^{-1} mod 26 $= 26 = 11(2) + 4$

$11 = 4(2) + 3$ $\qquad 4 = 4 - 3(1)$

$4 = 3(1) + 1$ $\qquad 1 = 4 - 3(11 - 4(2))$

$1 = (4) - (11)$

$1 = (26 - 11(2) - (11) = -6(11) - (11) + 3\times 26$

$1 = (-7)\,11 + 3\times 26$, $-7 + 26 = 19$

11^{-1} mod $= 19$.

$$19\begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} = \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \text{ mod } = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

8. Find the inverse of the following matrix, whose ethire are considerd mod 26

$\Rightarrow$ $\begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix} = (11\times 3) - (13\times 2) = 33 - 26 = 7$

7^{-1} mod 26 $\Rightarrow$ $26 = 7(3) + 5$

$7 = 5(1) + 2$

$5 = 2(2) + 1$

$\Rightarrow$ $1 = 5 - 2(2)$

$1 = 5 - 2(7 - 5)$

$1 = 3(5) - 2(7)$

M T W T F S S

COMPASS

Date :

2) $1 = 3(26 - 7(3)) - 2(7)$

$1 = (-11)\,7 + 3(26)$

$-11 + 26 = 15$

$7^{-1} \bmod 26 = 15$

$$15\begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix} = \begin{bmatrix} 45 & -195 \\ -30 & 165 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix}$$ //

9. Is the key valid ? $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$

Ans: $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix} = (2 \times 21) - (13 \times 14) = (2,2) - (182)$

$= 140$

140 ~~goes~~ is not coprime with 26.

10. Encrypt using 2x2 Hill cipher for the PT "SOLVED" using $\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$

Ans: for Encryption : $C = P * K \bmod 26$.

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = (3 \times 4) - (7 \times 8) = (33 - 56) = 23$$

23 is non-zero.

23 is coprime with 26.

plaintext : solved.

i) $P = \begin{bmatrix} S \\ O \end{bmatrix} = \begin{bmatrix} 18 \\ 14 \end{bmatrix}$

$$\begin{array}{cc} K & P \\ \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} & \begin{bmatrix} 18 \\ 14 \end{bmatrix} \end{array} = \begin{bmatrix} 3 \times 18 + 8 \times 4 \\ 7 \times 18 + 11 \times 14 \end{bmatrix} = \begin{bmatrix} 166 \\ 280 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 10 \\ 20 \end{bmatrix} = \begin{bmatrix} K \\ U \end{bmatrix}$$

ii) $P = \begin{bmatrix} L \\ V \end{bmatrix} = \begin{bmatrix} 11 \\ 21 \end{bmatrix}$

$$\begin{array}{cc} K & P \\ \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} & \begin{bmatrix} 11 \\ 21 \end{bmatrix} \end{array} = \begin{bmatrix} 3 \times 11 + 8 \times 21 \\ 7 \times 11 + 11 \times 21 \end{bmatrix} = \begin{bmatrix} 201 \\ 308 \end{bmatrix} \bmod 26$$

$$\begin{bmatrix} 19 \\ 22 \end{bmatrix} = \begin{bmatrix} T \\ W \end{bmatrix}$$

iii) $P = \begin{bmatrix} E \\ D \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$

$$\begin{array}{cc} K & P \\ \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} & \begin{bmatrix} 4 \\ 3 \end{bmatrix} \end{array} = \begin{bmatrix} 3 \times 4 + 8 \times 3 \\ 7 \times 4 + 11 \times 3 \end{bmatrix} = \begin{bmatrix} 36 \\ 61 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 10 \\ 9 \end{bmatrix} = \begin{bmatrix} K \\ J \end{bmatrix}$$

Cipher text = KU TWKJ